

Group 1
L04 Exploratory Data Analysis
ITAI-1371
Viswanatha Rao

Reflection Journal – Exploratory Data Analysis

Before this lab, our group mostly thought of Exploratory Data Analysis as a quick way to check a dataset before moving on to machine learning. After working with the Titanic data, we started to understand why this stage matters much more than we first thought. Looking at the first rows, data types, missing values, and summary statistics gave us a general picture of the dataset, while the visualizations showed patterns that were difficult to understand from numbers alone. For example, we found that only about 38.4% of the passengers survived, which showed us right away that the survival outcomes were not evenly balanced.

One of the biggest lessons from this lab was learning the difference between looking at statistics and actually understanding the patterns behind them. A mean or count can summarize hundreds of observations, but it does not always explain how different groups compare. The plots made those differences much clearer. The passenger-class visualization showed a large survival difference between first- and third-class passengers, while the gender visualization showed that females survived at a much higher rate than males. These results helped us understand that visualizations are an important part of exploring and interpreting data, not just something we add to make a notebook look better.

The experimentation section also changed the way we interpreted relationships in data. In the port-of-embarkation experiment, passengers who boarded at Cherbourg had more survivors than deaths, while Southampton and Queenstown showed more deaths than survivors. At first, it could be easy to conclude that boarding at Cherbourg directly improved someone's chance of survival. After looking at the data more closely, though, we realized that this explanation would be too simple. Passengers who boarded at Cherbourg might also have differed in passenger class, fare, gender, or other factors that affected survival. This helped us understand that EDA can show a connection between variables, but it cannot prove that one variable directly caused another.

The fare experiment led to a similar realization. Survivors generally paid a higher median fare and showed a wider range of fare values, and the boxplot also included several large outliers. However, fare also connects closely with passenger class, so we could not treat it as an independent explanation for survival. Seeing these relationships together made the workflow feel more complex, but also more logical. Different features can affect each other or represent similar information, so we learned that we need to look at how variables connect instead of studying each one by itself.

This lab connected well with what we learned about learning types and workflow in the previous lab. We understood supervised learning as a situation where the model has known labels

during training, while unsupervised learning searches for patterns without an answer key. Since the Titanic dataset already includes the Survived column, predicting whether a passenger survived would be a supervised learning problem. However, this lab showed us that having labels does not mean we should immediately train a model. We first need to understand missing values, feature distributions, class balance, and relationships in the data. Because of this, EDA now feels like an important part of the machine learning process instead of just a separate step that comes before it.

We also faced a practical challenge because some notebook cells did not run correctly when several lines had merged together. Fixing the issue took more than simply copying and pasting new code. We had to figure out what each line was supposed to do, put the code back into the correct structure, and run the cells again to make sure everything still worked properly. This made debugging part of the learning process and reminded us why checking the notebook from beginning to end matters.

Moreover, the lab left us with several questions. We would like to know how strongly the relationship between embarkation and survival remains after considering class and fare. We are also curious about the best way to handle missing age and cabin values before model training. Another question is how the larger number of passengers who died would affect model evaluation and whether accuracy alone could become misleading. These questions showed us that EDA does not always give one final answer. Instead, it often helps us figure out what we should explore next.

In conclusion, this lab changed how our group looks at the beginning of a machine learning project. We no longer see EDA as just a set of plots that we need to complete. Each graph helps us understand the data, question our first assumptions, and make better decisions about later steps. The biggest lesson was that a model can only learn from the information we give it, so understanding that information has to come before choosing an algorithm or judging its performance.